

MIKE HOLZINGER

mike.holzinger@gmail.com

Brooklyn, NY 11213

Senior Software Engineer with 18+ years across Python, C, and systems programming at IBM, NASA, Disney, Cisco, and ExodusPoint.

Experienced in code review, debugging, unit testing, and technical documentation. Daily practitioner of AI-assisted development workflows (Claude Code, MCP) including writing and evaluating LLM-generated code.

Skills and Expertise

Languages: Python (9 yrs), C (10 yrs), Ruby (2 yrs), Shell/Bash (18+ yrs)

Developer Tools & Automation: GitLab CI/CD, GitHub Actions, Jenkins, Claude Code, MCP (Model Context Protocol)

Containers & Orchestration: Docker, ArgoCD

Cloud Platforms: AWS (ECS, S3, SES), GCP (GKE), IBM Cloud

Observability & Monitoring: Grafana, Prometheus, ELK Stack, OpenSearch, OpenTelemetry

Security & Compliance: PKI, FIPS 140-2, Certificate Authority Management, AES Encryption, HashiCorp Vault, LDAP/Active Directory

Databases: MySQL/MariaDB, PostgreSQL, CouchDB, OpenSearch

Operating Systems: Linux/Unix Administration (18+ yrs), Windows Server Administration

EXODUS POINT CAPITAL MANAGEMENT

Sr. Software Engineer (Contract work - Completed)

April 2025 - May 2026

Developer Experience & Python Platform:

- Led Developer Experience initiative to design and deliver a custom Python environment distribution platform for quantitative analytics teams, reducing deployment time from 30 minutes to under 2 minutes

- Built in-house developer tooling using GitLab CI/CD and Windows automation, giving engineering teams full ownership and control of their Python environments without vendor dependencies
- Developed automated deployment tooling integrating with existing Windows infrastructure, enabling seamless environment activation and application integration for portfolio management workflows
- Served as primary Developer Experience contact and Python support lead for organization-wide environment standardization across quantitative analytics teams

Platform Engineering & Automation:

- Built and operated secure AWS infrastructure stack for an automated email processing pipeline, including encrypted bucket architecture and access-controlled data flows
- Developed Kubernetes service for real-time email monitoring and routing, integrating LDAP-based filtering to match incoming messages against Active Directory group membership
- Built Python filtering pipeline processing incoming email from secured source buckets into organized, structured output buckets sorted by user and date
- Integrated platform components with an isolated LLM summarization service and OpenSearch indexing layer, serving as the operations and data pipeline engineer for the end-to-end system

Key Technologies:

Python, PowerShell, GitLab CI/CD, Docker, AWS, OpenSearch, LDAP/Active Directory

CISCO

Bayone Solutions

Sr. Software Engineer (Contract work - Completed)

April 2024 - October 2024

WebEx Framework Modernization:

- Contributed to modernizing the WebEx corporate framework, configuring and deploying custom builds for over 300 clients, enhancing the platform's flexibility and performance
- Migrated the codebase from Python 2.8 to Python 3.12, utilizing tools like Tox and Pytest for testing and validation, ensuring code reliability and compatibility

Configuration Architecture & Automation:

- Enhanced the configuration architecture by implementing design patterns and Jinja templating, improving maintainability and scalability of configuration management
- Developed new linting tools to validate JSON, YAML, and environment key settings, ensuring consistency across more than 5,000 configuration documents.

- Streamlined the deployment process using Docker and Jenkins, reducing build times and improving continuous integration workflows

Collaboration & Testing:

- Collaborated with cross-functional teams to create comprehensive unit tests for system functionality, increasing test coverage and software quality

Key Technologies:

Docker, GitHub, Jenkins, Jinja2, Pytest, Python 3.x, Tox, JSON, YAML

NASA

MRI Technologies

Sr. Software Engineer

March 2023 - November 2023

CI/CD Pipeline Optimization:

- Refactored GitLab pipelines from Docker-in-Docker to Kaniko, reducing deployment times from 1 hour to under 8 minutes and eliminating legacy dependencies
- Automated CI processes through Python scripting, improving platform reliability and developer productivity for NASA's cloud platform

Developer Tooling & Automation:

- Built database automation tooling for job scheduling and export operations, contributing to the efficiency and reliability of NASA's cloud platform
- Applied Helm version locking to mitigate security vulnerabilities and improve dependency management

Developer Enablement:

- Deployed Grafana observability dashboards and trained development teams on metrics and logging workflows, giving developers visibility into deployed changes

Key Technologies:

Python, GitLab CI/CD, Docker, Grafana, Google Cloud Platform

Disney (Parks and Recreation)

HTC Global

Sr. Software Engineer - Automation & Integration (Contract - Completed)

September 2022 - April 2023

Python Automation & Integration:

- Built Python automation platform integrating ServiceNow with Slack Notification API, eliminating manual escalation processes and reducing missed change management windows
- Developed Python-based monitoring tooling using AppDynamics and Grafana APIs for proactive incident detection, contributing to increased application reliability
- Automated application onboarding workflows for large-scale migration affecting 500+ applications

Key Technologies:

Python, AppDynamics, Grafana, Prometheus, ServiceNow, Slack Workflow Builder, AWS ECS

SMARSH

Software Engineer - Automation & Tooling

July 2021 - September 2022

Automation & Developer Tooling:

- Built automation tooling for mutual TLS configuration and OpenTelemetry application instrumentation across distributed services
- Developed configuration automation for SOLR private cloud and AMQ administration tooling
- Migrated observability stack from Grafana to Datadog using OpenTelemetry as data pipeline, improving developer visibility into system performance
- Automated CI/CD pipelines using CircleCI and GitHub Actions, reducing manual deployment overhead across engineering teams

Key Technologies:

Python, Ansible, Docker, GitHub Actions, HashiCorp Vault, OpenTelemetry, PowerShell

INDEPENDENT SOFTWARE DEVELOPMENT

Software Engineer - Independent Projects

November 2019 - June 2021

- Contributed to open-source projects in the MiSTer FPGA retro computing ecosystem
- Built ROM curation and metadata automation tooling in Python and Bash for game preservation communities
- Designed and performed embedded hardware modifications for Game Boy consoles, including custom component-level builds

IBM

Software Engineer

July 2015 - October 2019

Developer Tooling (**Watson Data Platform / Cloudant**):

- Built internal support tooling that eliminated direct server node access for operators, reducing investigation time and improving developer self-service capabilities
- Developed cluster health investigation tools aggregating multi-node data into unified diagnostic views
- Wrote metrics performance tooling for FoundationDB, laying groundwork for next-generation document store infrastructure on the Watson Data Platform

Automation & Security Tooling (**Global Managed Cloud / GBS**):

- Built automated host discovery system for compliance reporting, eliminating manual inventory processes across distributed infrastructure
- Designed and built compliance reconciliation service automating previously manual data remediation workflows
- Automated GitHub org-wide two-factor authentication enforcement by monkey-patching the Octokit Ruby client to enable a beta API capability during the Microsoft acquisition period
- Integrated GitHub API controls for dual-control code review enforcement and automated metrics collection for real-time inventory and auditing

Platform Tooling (**IBM Cloud / Object Storage / OpenStack**):

- Built automated inventory deployment and account provisioning tooling for petabyte-scale IBM Cloud S3 Object Storage deployments
- Developed client auditing and account remediation tooling for Cleversafe acquisition integration

Developer Enablement (**West Coast Operations Lead**):

- Led onboarding and technical training for new engineers and operations staff across IBM and BlueBox transfers
- Built Grafana and Sensu monitoring dashboards for operational visibility across OpenStack deployments

Key Technologies:

Python, Ruby, BASH/Shell, REST APIs, GitHub Enterprise, Chef, Ansible, CouchDB, FoundationDB, Grafana, Linux

BLUE BOX GROUP

Software Engineer - Linux & Security

July 2014 - July 2015

Security & Certificate Authority:

- Served as Registrar for the published Certificate Authority (CA) and managed the internal CA, ensuring secure communications and compliance with security standards
- Maintained the corporate bastion/authentication gateway, enhancing security protocols for internal systems
- Implemented SSH key authentication and YubiKey/Yubico hardware security for access management

Database & Infrastructure:

- Provided admin and on-call support for MySQL/MariaDB and PostgreSQL instances
- Built new primary and secondary SQL solutions and implemented disaster recovery strategies to ensure data integrity and availability
- Key team member on migration of customer tenant OpenVZ applications, databases, and infrastructure to OpenStack

Key Technologies: Python, Ruby, Perl, PKI, OpenSSL, MySQL/MariaDB, PostgreSQL, Grafana, Nagios, HAProxy, Nginx, YubiKey/Yubico

TOWNSEND SECURITY

Software Engineer

2007 - 2013

Cryptographic Client Library Development:

- Utilized **ANSI C** to develop lightweight client libraries for the **Alliance Key Manager** (Hardware Security Module).
- Created a **TLS client library in C** for partners to interface with the key manager, ensuring secure and efficient communication.
- Focused on clean code practices in config file parsing, TLS connection management using FIPS-compliant OpenSSL modules, and certificate generation.

FIPS 140-2 Certification:

- Contributed to the development of cryptographic libraries that achieved **NIST FIPS 140-2 certification**, ensuring compliance with stringent security standards.
- Implemented AES encryption, PRNG, SHA, and HMAC algorithms in accordance with NIST guidelines.

Cross-Platform Development & Automation:

- Developed cross-platform SaaS client applications compatible with both Windows and Linux environments.
- Utilized **Jenkins** for continuous integration, automating builds triggered by Git commits and

creating testing harnesses to validate new builds.

- Built a central **Certificate Authority** for testing, demos, and partner deployments using **Bash** and **OpenSSL**.
- Created metrics testing tools for encryption API IOPS, high availability failover, and scaling on AWS using **Perl** and **Bash**.
- Developed a consumer PKI deployment tool for SSL certificates on Windows/Linux platforms using **OpenSSL** and **Nullsoft**

Key Technologies:

ANSI C, AES, PKI, TLS, FIPS 140-2, OpenSSL, Jenkins, Bash/Shell, Perl, Git, Linux

EDUCATION

Green River College

Associate of Arts